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These points are named for a horseâ€™s saddle. When youâ€™re sitting on a horse, youâ€™re at a point that is flat ($\nabla f(\mathbf{a}) = \mathbf{0}$), but the saddle goes up in one direction (toward the horseâ€™s head) and down in another direction (towards the sides).

Summarizing the results from the Second Derivative Test:

Corollary twice cont. diff'le.

Suppose that f is C^2 and $\nabla f(\mathbf{a}) = \mathbf{0}$.

$\nabla^2 f(a)$

- *If $H(\mathbf{a})$ is positive definite, then $\mathbf{a}$ is a local minimum point;*
- *If $H(\mathbf{a})$ is negative definite, then $\mathbf{a}$ is a local maximum point;*
- *If $H(\mathbf{a})$ is indefinite, then $\mathbf{a}$ is a saddle point, by definition.*
- *If none of the above hold, then we cannot determine the type of critical point by this test.*

Def. 5 (Degenerate critical point)
If f is a C^2 function and $\mathbf{a}$ is a critical point of f, we say that a critical point is *degenerate* if $\det H(\mathbf{a}) = 0$, and *nondegenerate* if $\det H(\mathbf{a}) \neq 0$.

determinant of a matrix.

If a critical point is nondegenerate, then the Corollary says that by finding all the eigenvalues of the Hessian, we can immediately determine whether it is a local max, local

Ex $f(x,y) = x^2 + y^2 \to$ convex

$\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x = 0 \\ y = 0 \end{matrix}$ } stationary pt. $\downarrow$ global min

Ex $f(x,y) = x^2 + 2xy + y^2 \to$ convex.

$\nabla f = \begin{bmatrix} 2x + 2y \\ 2x + 2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$x = -y$, $y = -x$ $\to (-a, a)$ local. global. $a \in \mathbb{R}$.

$\to \nabla^2 f = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ Find eigenvalue: $\det \begin{bmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{bmatrix} = 0$

$|\nabla^2 f - \lambda I| = 0$

$(2-\lambda)^2 - 4 = 0$

$2 - \lambda = \mp 2$

$2 - \lambda = 2$, $\lambda = 0$ ✓ $\quad 2 - \lambda = -2$, $4 = \lambda$

positive semi-definite

$\boxed{\lambda_i \geq 0}$

$f(-a,a) = a^2 - 2a^2 + a^2 = 0 < f(x)$? $x \in N$

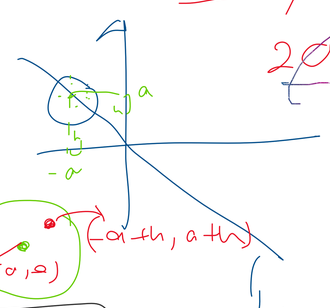
$a > 0$

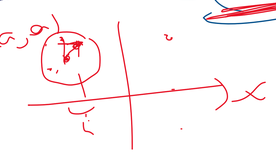
from neigh.

$f(\underbrace{-a+h, a+h}_{\in N(-a,a)}) = (-a+h)^2 + 2(-a+h)(a+h) + (a+h)^2$

$a^2 - 2ah + h^2 - 2(h^2 - a^2) + a^2 + h^2 + 2ah$

$2a^2 + 2a^2 \to = 4a^2 \neq 0$.





$y = -x$

$f(-a,a) \leq f(x)$ $\quad x \in N$

$\Leftrightarrow 0 \leq 4a^2$

$f(-a,a) \leq f(x)$

$\forall x \in N$ local min. $\Downarrow$ convex global min.

Example 4.

Let $f(x,y) = x^2 + ay^3 + by^4$. Show that the origin is a critical point and classify it. The answer will depend on a and b.

▾ **Solution**

We first compute $\nabla f = (2x, 3ay^2 + 4by^3)$, so the origin is a critical point. To classify it, we need the Hessian

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 6ay + 12by^2 \end{pmatrix}.$$

Thus $H = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ at the origin, for all a and b, with eigenvalues 0 and 2. Thus we cannot determine the type of critical point from the second derivative test. We can however say that it is neither a local maximum nor a saddle point.

So the critical point is either a local minimum, or â€œnone of the aboveâ€□. It is a local minimum only if $f(x,y) \geq 0$ for all (x,y) close to $(0,0)$. The points $(0,\varepsilon)$ have values approximately $a\varepsilon^3$ when ε is small, so they can be made negative if $a \neq 0$. When $a = 0$, they have value $b\varepsilon^4$. Hence, it is a local minimum if $a = 0$ and $b \geq 0$, and otherwise it is neither a saddle point nor a local extremum.